Suggesting Categories for Wikimedia Commons Images

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Abstract

We propose a tool that helps Wikimedia Commons volunteers categorize new images. It recommends relevant categories for human review rather than editing files automatically. By combining image similarity, category descriptions, a supervised classifier, and upload text, the tool can suggest both common and less frequent categories while showing evidence for its recommendations.

1 Problem and Approach

Commons receives new images every day, but many remain uncategorized or are assigned categories that are too broad or not quite right. Categorization is also nuanced: an image may fit several categories, and editors often choose a specific category without adding its broader ancestors. As a result, a category missing from an image's labels is not necessarily an incorrect category. Meanwhile, most categories have few examples, and new ones continue to appear.

We will treat category assignment as a multi-label recommendation task with incomplete labels. Our system will offer suggestions to editors, who remain in control of accepting or rejecting them. This human-review step is important both for quality and for learning from feedback.

2 Proposed System

We will build an ensemble of four complementary scorers using one frozen, CLIP-style vision-language encoder. Each image is encoded once, keeping the system practical to run. Rather than limiting recommendations to a fixed list of model outputs, categories can also be retrieved by their similarity to images or category names. This makes the system easier to update as Commons grows.

Our four scorers are:

- **Image retrieval:** find similar training images with FAISS and use their categories to rank suggestions. These neighbours also give editors an explanation for each recommendation.
- **Category text matching:** compare the image with text embeddings of category names, providing a zero-shot signal that can help categories with few examples.
- **Supervised prediction:** train a linear, multi-label classifier with one sigmoid output per category and an asymmetric loss to account for rare positives and incomplete labels.
- **Upload text:** use TF–IDF and logistic regression on titles and descriptions, which can be particularly helpful for events and places.

We will combine the top candidates from these scorers using logistic-regression stacking trained on the validation set, and compare it with a simple weighted average. Editors will see ranked categories, confidence scores, and supporting neighbour images, with options to accept or reject suggestions. New images can be added to the retrieval index, and new categories can be introduced through their text embeddings without retraining the encoder. Review feedback will be saved as additional labels.

3 Data and Evaluation

We will collect about [N] openly licensed images across about [C] categories from 30 seed topics spanning objects, scenes, events, and abstract concepts. Using the MediaWiki API, we will retain thumbnails, upload dates, uploader names, descriptions, and visible categories. Maintenance and hidden categories will be removed, as will categories that are too rare for the initial experiment.

To test on genuinely new uploads, we will split images by upload date: the oldest 70% for training, the next 15% for validation, and the newest 15% for testing. We will also remove near-duplicates across splits using embedding similarity to reduce data leakage.

We will report per-image Precision@k, Recall@k, and F1@k for $k = 1, 3, 5, 10$, along with category-level macro averages and mean average precision (mAP). Because images can have several valid labels, top-1 accuracy alone would not tell the whole story. We will compare each scorer on its own with the ensemble, and compare stacking with weighted averaging. Results will be broken down by frequent, medium-frequency, and rare categories, as well as by topic type. We will also test robustness to blur, JPEG compression, and downscaling; measure latency and index size; and show that adding images or categories requires no encoder retraining.

Finally, labels are incomplete, so some apparently incorrect suggestions may in fact be valid. We will manually review a sample of these cases and report both strict scores and scores adjusted for the reviewed labels. Our dataset represents only a sample of Commons, and its existing labels may reflect geographic or cultural biases; these limitations will guide how we interpret the results.